Machine Learning Project

Heart Disease Prediction Using Machine Learning

Mini Project · Machine Learning Course ·

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Problem & Dataset

Understanding the critical challenge and data foundation

The Problem

17.9M deaths / year

Cardiovascular disease is the world's leading cause of death

"Silent Ischemia"

Patients feel **NO symptoms** yet have serious disease

Missed by Triage

Symptom-based screening misses the highest-risk patients

Solution: Automated ML-based screening to catch silent cases

Dataset Summary

Source	Kaggle — fedesoriano, 2021
Patients	918
Features	11
Target	HeartDisease (0/1)

Class Distribution

Disease	Healthy
508	410
55.3%	44.7%

Features include age, chest pain type, cholesterol, exercise test results

Data Cleaning & Preprocessing

5-step pipeline ensuring data quality and model readiness

1 Handle Missing Values

172 Cholesterol zeros

Biologically impossible → missing values

Replaced with **median** (237 mg/dl)

Median chosen over mean to avoid outlier bias

2 Remove Duplicates

```
df.drop_duplicates()
```

0 duplicates found

Step always performed as best practice to avoid training bias

3 Encode Categorical Variables

Sex	ChestPainType	RestingECG	ExerciseAngina	ST_Slope

LabelEncoder applied: Converts text → integers for ML algorithms

4 Train/Test Split

80%
Train
734 patients

20%
Test
184 patients

stratify=y preserves 55/45 class ratio

5 Feature Scaling

StandardScaler

mean ≈ 0, std ≈ 1

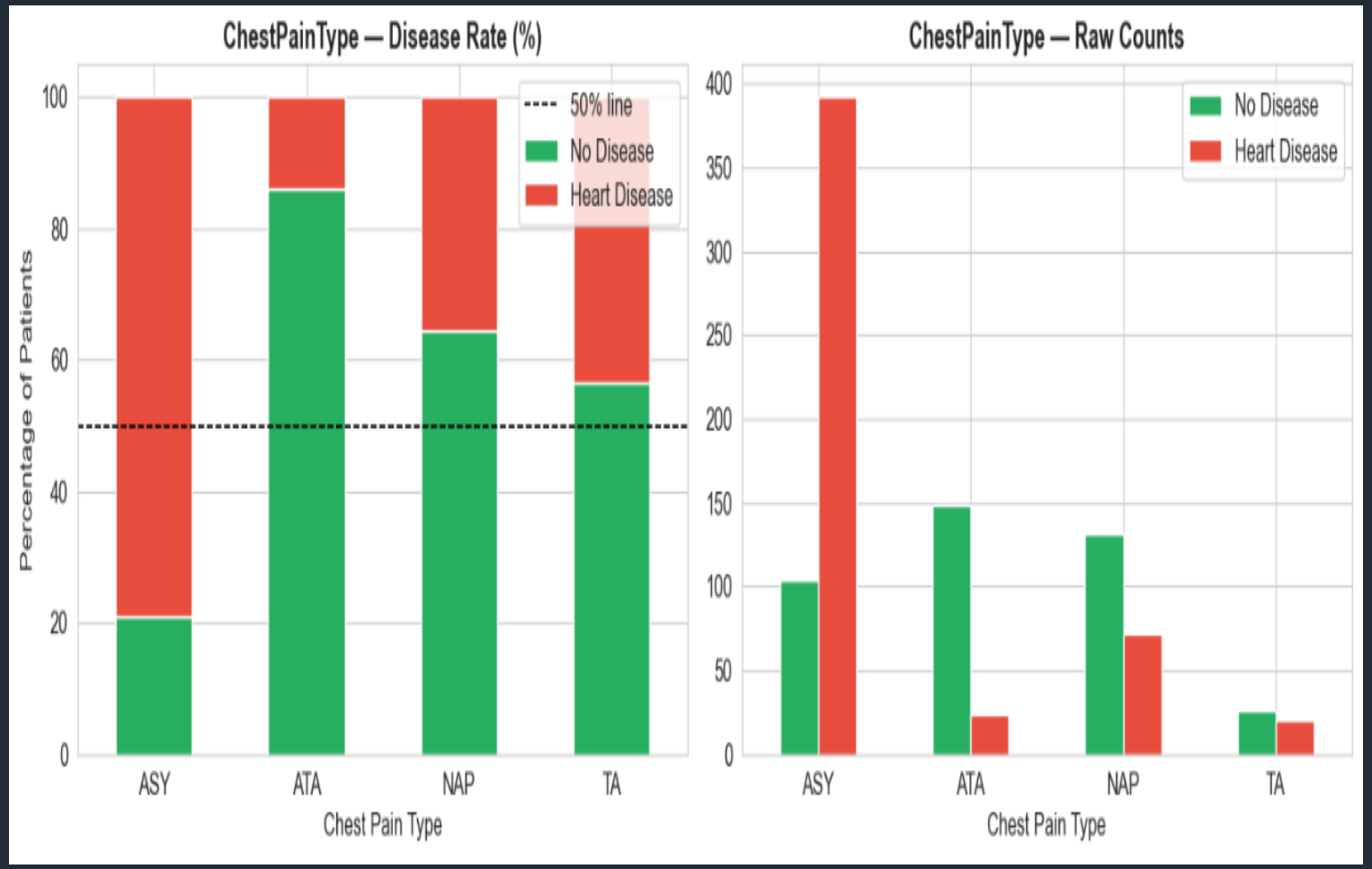
Critical: fit_transform on TRAIN only

Applying to test = data leakage

EDA — Key Insight #1: Silent Ischemia

The most critical finding: asymptomatic patients have the highest disease rate

Chest Pain Type vs Heart Disease



79% Disease Rate

for ASY patients

ASY (Asymptomatic) patients feel **NO chest pain** yet have the HIGHEST disease rate

Key Insight

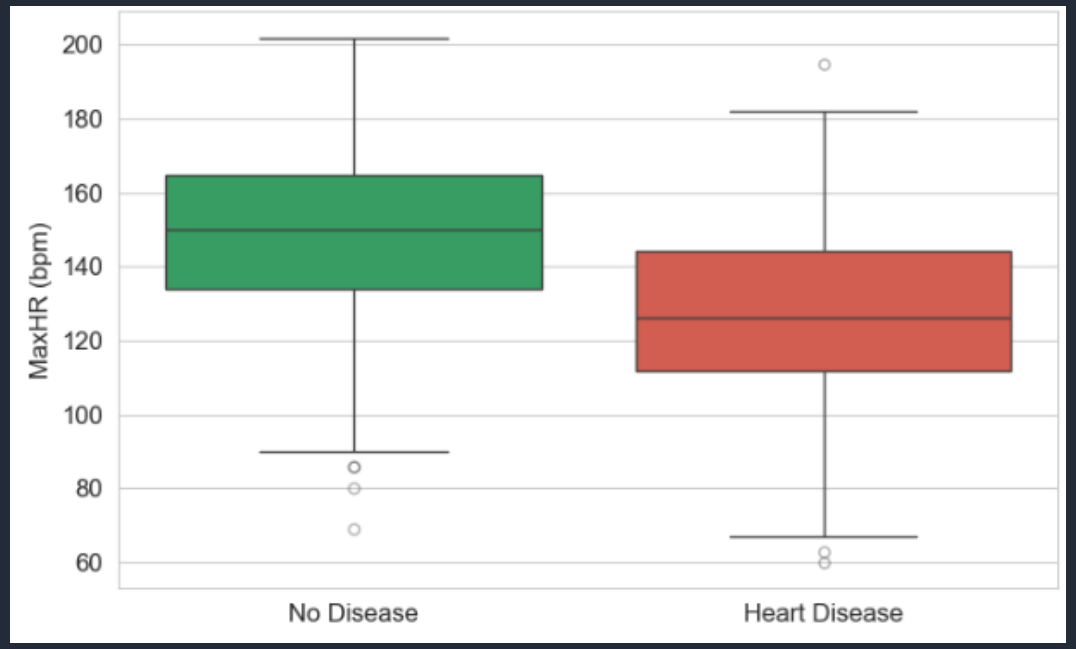
- This is "silent ischemia" — completely missed by symptom-based triage
- ML screening is essential to catch patients who feel fine but are at highest risk

Clinical Significance: Traditional screening fails the most vulnerable patients

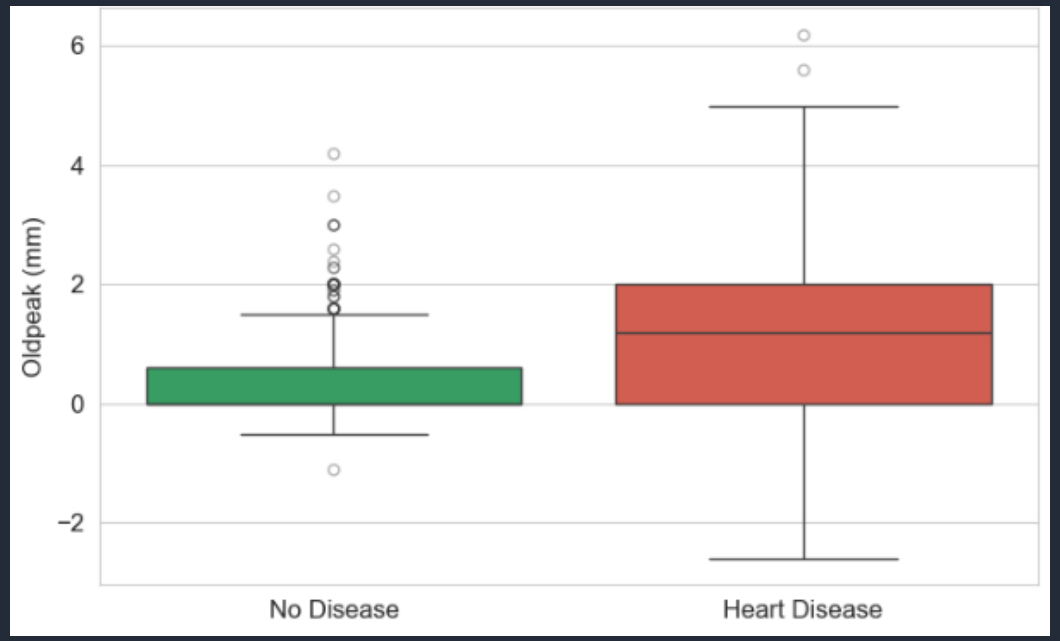
EDA — Key Insight #2: Exercise Test Features

Clear separation between disease and healthy patients in exercise stress test metrics

Maximum Heart Rate (MaxHR)



ST Depression (Oldpeak)



Lower MaxHR

148 → 127 bpm

Healthy → Disease

Higher Oldpeak

0 → 1 mm

Healthy → Disease

Clinical Interpretation

A sick heart cannot perform under exercise stress — these features became the most important predictors

Models Tested

5 algorithms from 4 different ML families

Logistic Regression

Linear Model

Finds a straight-line boundary between classes

Simple | Fast | Interpretable

Used as our **LINEAR BASELINE**

Decision Tree

Tree-Based

Asks a series of yes/no questions (e.g., Age > 50?)

Readable Rules | Overfitting Risk

Creates **if-then rules**

SVM

Margin-Based

Finds the widest possible gap (margin) between classes

RBF Kernel | Non-linear

Handles **non-linear boundaries**

Random Forest

★ PRIMARY

Ensemble — Bagging

Trains 200 trees on random data subsets. Majority vote of all trees.

200 Trees | Reduces Variance

★ **Primary Model**

Gradient Boosting

Ensemble — Boosting

Builds trees one-by-one, each correcting the previous one's mistakes

Sequential | LR=0.05

Small learning rate = **robust**

Methodology: Test diverse algorithm families

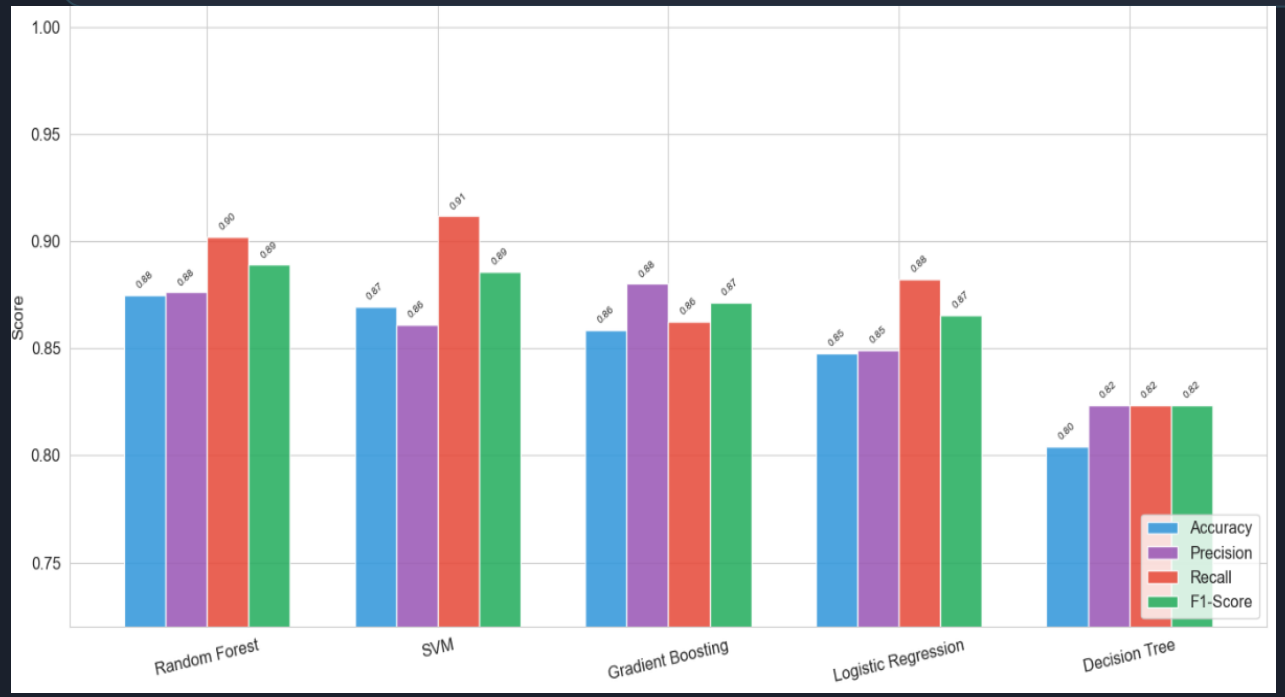
Goal: Compare performance across approaches

Results Comparison

Comprehensive performance evaluation across all metrics

Model Performance Metrics

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Random Forest	**87.5%**	87.6%	**90.2%**	**88.9%**	**0.923**
SVM	87.0%	86.1%	**91.2%**	88.6%	0.919
Gradient Boosting	85.9%	88.0%	86.3%	87.1%	**0.925**
Logistic Regression	84.8%	84.9%	88.2%	86.5%	0.899
Decision Tree	80.4%	82.4%	82.4%	82.4%	0.863



Best Model

Random Forest

F1-Score
88.9%

Recall
90.2%

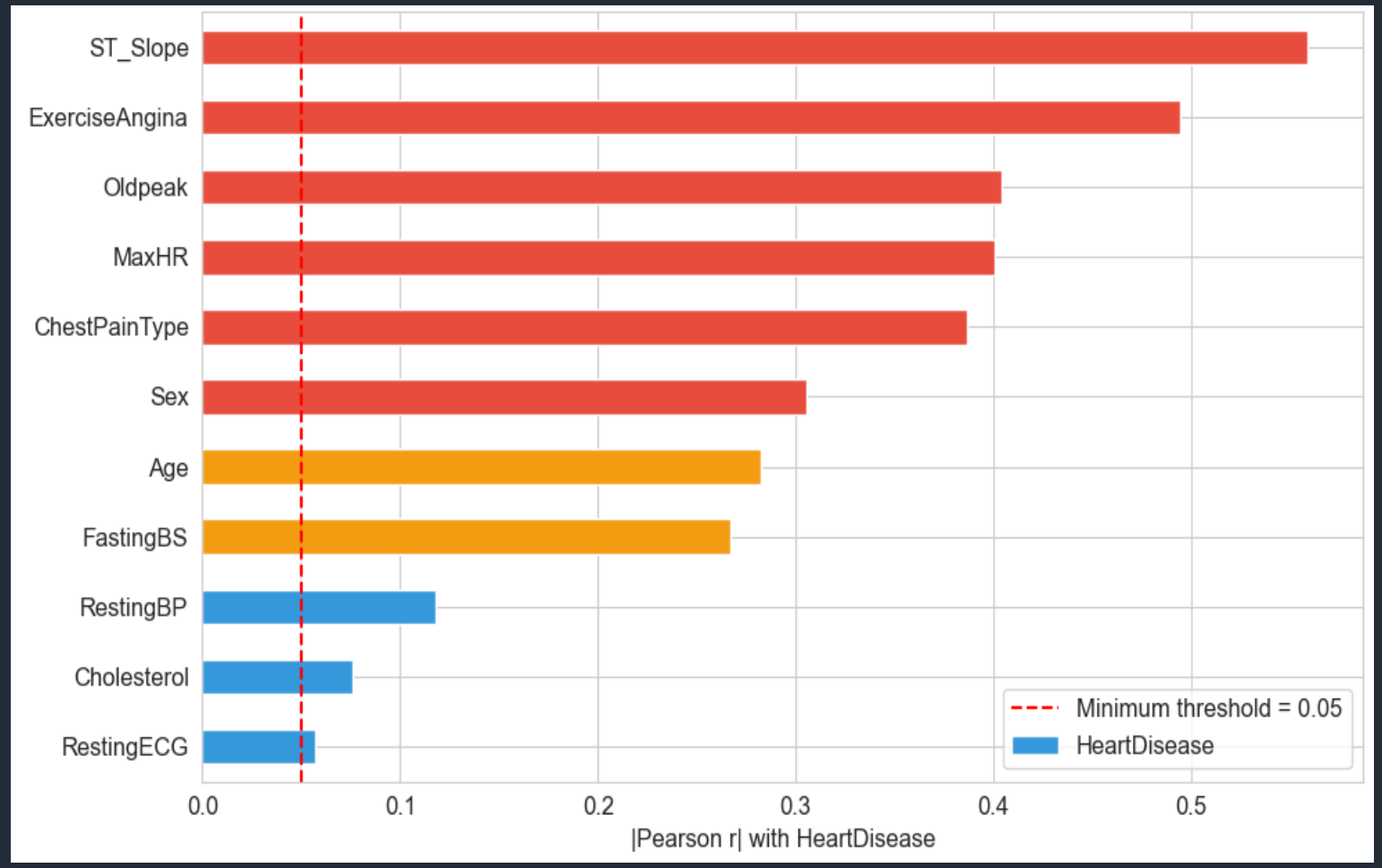
Why Random Forest?

- Highest F1-Score (88.9%)
- Best Recall (90.2%)
- Stable cross-validation scores

Feature Importance & Explainability

Understanding what the model learned and its clinical validity

Feature Relevance — Correlation with Target



Clinical Interpretation

1. **ST_Slope**
   ECG slope at peak exercise — **strongest ischemia marker**
2. **ExerciseAngina**
   Chest pain during exercise = **coronary supply failure**
3. **Oldpeak**
   ST depression **quantifies myocardial stress**
4. **MaxHR**
   Sick heart **can't sustain high rates** under stress

Validation: Matches clinical cardiology guidelines — exercise ECG is the gold standard

Conclusion

Key achievements, clinical validity, and real-world impact

Complete Pipeline

Data Cleaning → Pre-processing → EDA

↓

Modeling → Evaluation

Best Model

Random Forest

F1-Score	Recall
88.9%	90.2%

Catches 9 out of 10 sick patients

Clinical Validity

Top features all from exercise ECG:

- ✓ ST_Slope
- ✓ ExerciseAngina
- ✓ Oldpeak
- ✓ MaxHR

Matches **medical guidelines**

Real-World Impact

Detects silent ischemia patients who feel no pain

⚠ Exactly the group traditional screening MISSES

Thank You!

Questions?